The Cayley Transform in Physics:
half-planes, disks, and half-angles

An expository note

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Abstract

The Möbius map $w = (z - i)/(z + i)$ turns up across physics and engineering under many names. It is the half-plane-to-disk map for Gaussian beams and squeezed states, the conversion from quadratures to sidebands, the Smith chart, Tustin's bilinear transform, Crank–Nicolson time stepping, Heitler's reaction matrix, von Neumann's treatment of self-adjoint operators, and the change from linear to circular polarization. Wherever it appears, an angle on one side becomes twice an angle on the other. This note collects these appearances, explains what they share, and traces them to the literature.

The map does two different jobs. Sometimes it changes coordinates on a space of *states* (beams, squeezed states, loads). Sometimes it acts on a *generator* or spectral variable (frequency, energy, a Hamiltonian, a time step). Behind both jobs lies one structure. A quadratic quantity of signature $(1, 1)$ (a symplectic form, a flux, a Wronskian, a Lagrange invariant) is skew in a real basis and becomes a difference of norms in a basis of conjugate pairs. The Cayley matrix is exactly the change from one basis to the other, and it carries the lossless boundary of one description onto that of the other. The doubled angle has one cause. The quantities physicists measure are ratios of the two members of a conjugate pair, and the two members rotate in opposite directions. We separate results found in the literature from the few observations we did not find stated anywhere.

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1 Three appearances of one map

This note began with three places where the same map turned up.

1. Gaussian beams. A paraxial Gaussian beam is described by a complex beam parameter q. Physical beams fill a half-plane, and a lossless optical system with ray matrix $\left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right)$, $AD-BC=1$, acts by the Möbius map $q \mapsto (Aq+B)/(Cq+D)$ [1]. A companion note [2] explains why: a Gaussian beam is a *complex ray*. The space of beams is the hyperbolic plane, optical systems act on it as isometries, and mode matching is a hyperbolic distance. Mode matching and resonator stability are easiest to see in the *disk* model of that plane, and the half-plane is carried to the disk by the Cayley map.

2. Quadrature directions. In the two-photon formalism of gravitational-wave interferometry [3, 4], a homodyne readout picks a direction $u = (u_1, u_2)$ in the plane of quadratures (a_1, a_2). The direction matters only up to sign. A second companion note [5] encodes it by the "double-angle phasor"

$$\chi(u) = \frac{u_1 + iu_2}{u_1 - iu_2}, \qquad |\chi| = 1, \qquad \chi(R(\alpha)u) = e^{2i\alpha}\chi(u), \tag{1}$$

which turns the design of frequency-dependent readout filters into placing the poles of a rational all-pass function.

3. Discretizing time. To turn a continuous-time controller into a digital one, engineers substitute $z = (1 + sT/2)/(1 - sT/2)$ [6]. Physical frequencies $s = i\Omega$ land on the unit circle at angle $\Omega_d T = 2\arctan(\Omega T/2)$.

All three are the map $w = (z - i)/(z + i)$, up to a rotation, and all three show the same factor of two: χ rotates by 2α, and the Tustin angle is *twice* an arctangent. This note asks three questions:

1. Is it the same structure each time?
2. Why does the angle double?
3. Where else does it appear?

The answers are partly in the literature, but scattered across optics, network theory, control, nuclear physics, random-matrix theory and operator theory. We found no source that ties them together. The closest is the review by Sánchez-Soto *et al.* [7], which treats transfer matrices in several branches of physics and says of the half-plane–disk map: "This remarkable map is known as the Cayley transform."

Plan.

- Section 2 sets out the mathematics once: the scalar map, its matrix, its action on Lie algebras, the two roles it plays, and the reason for the doubled angle.
- Sections 3–4 treat the anchors.
- Section 5 treats impedance and scattering, where both roles appear at once and where the KLMTV filter design turns out to be a reaction-matrix problem.
- Section 6 collects further examples.
- Section 7 draws the conclusions.

Every displayed formula with numerical content was checked numerically (Appendix B).

Conventions. We use the physics time convention $e^{-i\omega t}$, except in §4 and §5.1, where engineering sources use $e^{j\omega t}$ and $s = \sigma + j\omega$. Every source we read uses some rotated or reflected version of the Cayley map. Appendix A translates between them. These differences are always a rotation of the disk or of the half-plane, never anything deeper. Most of the work in combining sources is bookkeeping of this kind.

2 The map

2.1 Scalar form and the half-angle

The *Cayley transform*

$$\mathcal{C}(z) = \frac{z - i}{z + i}, \qquad \mathcal{C}^{-1}(w) = i\,\frac{1 + w}{1 - w}, \tag{2}$$

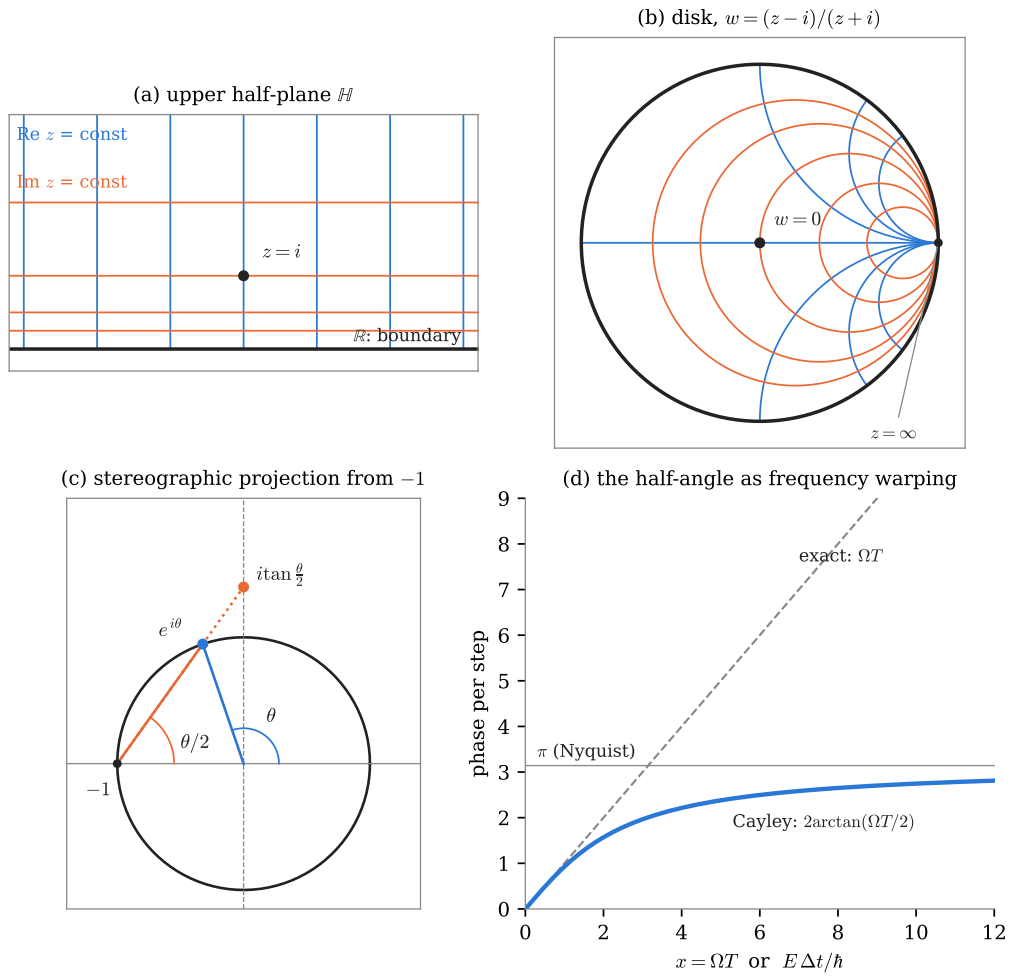


Figure 1: (a,b) The Cayley map (2) carries the coordinate grid of $\mathbb{H}$ to the disk. Vertical lines become circles through $w = 1$ (the image of $z = \infty$). Horizontal lines become horocycles tangent there. The boundary $\mathbb{R}$ becomes the unit circle. (c) The half-angle (4): stereographic projection from -1 sends $e^{i\theta}$ to $i\tan(\theta/2)$, by the inscribed-angle theorem. (d) The resulting warping $\theta = 2\arctan(x/2)$ against the exact $\theta = x$, as in Tustin's frequency warping (§4) and the phase error of Crank–Nicolson (§4.3).

sends the upper half-plane $\mathbb{H} = \{\operatorname{Im} z > 0\}$ onto the unit disk $\mathbb{D}$. It sends the real line onto the unit circle and the point i to the centre (Fig. 1a,b). Two identities make it useful in physics. The first measures how far a point is from the boundary on each side:

$$1 - |\mathcal{C}(z)|^2 = \frac{4 \operatorname{Im} z}{|z+i|^2}. \tag{3}$$

Positivity of $\operatorname{Im} z$ on one side is therefore exactly contraction $|w| < 1$ on the other. The second identity is what the real line looks like after the map:

$$\frac{1+ix}{1-ix} = e^{2i \arctan x}, \qquad \text{equivalently} \qquad e^{i\theta} = \frac{1 + i\tan(\theta/2)}{1 - i\tan(\theta/2)}. \tag{4}$$

The left side of (4) is $-\mathcal{C}(x)$, a rotation of (2) by π. Geometrically it is stereographic projection of the circle from the point -1 onto the imaginary axis (Fig. 1c). The line from -1 to $e^{i\theta}$ meets that axis at height $\tan(\theta/2)$, because an inscribed angle is half the central angle on the same arc. So the real coordinate x is the tangent of *half* the angle on the circle.

2.2 Matrix form: changing to a basis of conjugate pairs

As a matrix acting on homogeneous coordinates, (2) is $\left(\begin{smallmatrix} 1 & -i \\ 1 & i \end{smallmatrix}\right)$, up to scale. Physicists more often meet the normalized matrix with the rows in the other order,

$$\Omega = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & i \\ 1 & -i \end{pmatrix}, \qquad \Omega \begin{pmatrix} x \\ p \end{pmatrix} = \begin{pmatrix} a \\ a^\dagger \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} x + ip \\ x - ip \end{pmatrix}. \tag{5}$$

This is the passage from position and momentum quadratures to ladder operators [8]. It turns a real pair into a *complex-conjugate pair*. Conjugating by it maps $\mathrm{SL}(2,\mathbb{R})$ onto

$$\Omega\, \mathrm{SL}(2,\mathbb{R})\, \Omega^{-1} = \mathrm{SU}(1,1) = \left\{ \begin{pmatrix} \alpha & \beta \\ \bar\beta & \bar\alpha \end{pmatrix} : |\alpha|^2 - |\beta|^2 = 1 \right\}, \tag{6}$$

and it makes rotations diagonal [8, 9]:

$$\Omega \begin{pmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{pmatrix} \Omega^{-1} = \begin{pmatrix} e^{i\alpha} & 0 \\ 0 & e^{-i\alpha} \end{pmatrix}. \tag{7}$$

Apply Ω to a real direction $u = (u_1, u_2)$ and take the ratio of the two components. The result is (1):

$$\chi(u) = \frac{u_1 + iu_2}{u_1 - iu_2} = \frac{t+i}{t-i} = \frac{1}{\mathcal{C}(t)}, \qquad t = \frac{u_1}{u_2}. \tag{8}$$

The double-angle phasor is the Cayley transform of the projective coordinate of a direction.

Why conjugate pairs? The reason is a one-line identity, which Bonneau, Faraut and Valent use as the core of their treatment of self-adjoint extensions [10, eq. 29]:

$$|x + iy|^2 - |x - iy|^2 = 2i\,(\bar x y - x \bar y). \tag{9}$$

The right side is (up to a factor) an antisymmetric bilinear form in (x, y): a symplectic form, a Wronskian, a probability current, a Lagrange invariant. The left side is a difference of two squared norms. So a map that preserves the skew form in the real basis must preserve $|a|^2 - |b|^2$

in the conjugate basis. That is why $\mathrm{SL}(2,\mathbb{R})$ becomes $\mathrm{SU}(1,1)$, and why "lossless" becomes "unitary."

The same construction works for matrices. The Siegel upper half-space of complex symmetric $g \times g$ matrices with $\operatorname{Im} Z > 0$ is carried to the matrix disk by $W = (Z - i)(Z + i)^{-1}$, and the block version of Ω conjugates $\mathrm{Sp}(2g,\mathbb{R})$ into $\mathrm{SU}(g,g)$ [11–13].

2.3 From Lie algebra to group: Cayley's original problem

Cayley's 1846 paper solved a different problem: parametrize orthogonal matrices *rationally* [14]. In modern notation, if S is skew-symmetric then

$$\operatorname{cay}(S) = (I + S)(I - S)^{-1} \tag{10}$$

is orthogonal. For $n = 3$, Cayley wrote the parameters as $\tan(\frac{1}{2}\theta)$ times the direction cosines of the rotation axis. He credited them to Rodrigues [15] and noted their link with Hamilton's quaternions. The same parameters are Gibbs's "vector semi-tangent of version" $\mathbf{g} = \hat{n}\tan(\theta/2)$, with $\Phi = (I + I \times \mathbf{g}) \cdot (I - I \times \mathbf{g})^{-1}$ [16, Art. 127]. Gibbs also gives the cleanest geometric picture: the rotation carries $\mathbf{c} - \mathbf{g} \times \mathbf{c}$ to $\mathbf{c} + \mathbf{g} \times \mathbf{c}$. Read this as "half a step backward equals half a step forward." So the half-angle was already in the founding paper.

Hermite (1854) and Cayley (1858) went further. Their result covers any group preserving a bilinear form, $\{M : M^T \Omega M = \Omega\}$ [17, 18]. Cayley's 1858 memoir writes the transformation as $\Omega^{-1}(\Omega - T)(\Omega + T)^{-1}\Omega$, which is $(I - A)(I + A)^{-1}$ with $A = \Omega^{-1}T$. It states: "If Ω is a skew symmetrical matrix, then T is an arbitrary symmetrical matrix." In modern terms this is the statement that *the Cayley transform of a Hamiltonian matrix is symplectic.* The proof is one line:

Proposition 1. *If $A^T\Omega + \Omega A = 0$ and $I - A$ is invertible, then $M = (I + A)(I - A)^{-1}$ satisfies $M^T \Omega M = \Omega$.*

Proof. From $(I \mp A^T)\Omega = \Omega(I \pm A)$ we get $M^T\Omega M = (I - A^T)^{-1}(I + A^T)\Omega(I + A)(I - A)^{-1} = (I - A^T)^{-1}\Omega(I - A)(I + A)(I - A)^{-1} = (I - A^T)^{-1}\Omega(I + A) = \Omega$. □

The cases are:

- $\Omega = I$: skew $\to$ orthogonal;
- $\Omega = \operatorname{diag}(1, 1, 1, -1)$: Lorentz;
- $\Omega = J$: Hamiltonian $\to$ symplectic;
- the complex analogue: anti-Hermitian $\to$ unitary.

Iserles calls these *quadratic Lie groups* [19]. The exponential map also takes the algebra into the group, but the Cayley map has three advantages:

- It is rational.
- It is the $(1, 1)$ Padé approximant of e^{2A}, so $\operatorname{cay}(A/2) = e^A + O(A^3)$.
- It preserves the group because $\operatorname{cay}(-A) = \operatorname{cay}(A)^{-1}$.

Higher diagonal Padé approximants are products of Cayley factors and preserve the group too [19, 20]. Lewis and Simo used it directly in place of the exponential as a structure-preserving integrator [19, 21]. A modern statement is in [22]; the textbook home is [23].

Two cautions.

1. The map misses group elements with eigenvalue -1. Half-turns sit at infinity in Cayley coordinates. Every rotation is, however, the *square* of a Cayley transform, again by halving angles [24, 25].

2. Unlike exp, the Cayley map does not commute with taking representations or tensor products [26]. Take the spin-j generator $\hat{n} \cdot \mathbf{J}$ with $j \geq 1$. Its Cayley transform has eigenvalues $e^{2i \arctan(2\alpha m)}$, which are not of the form $e^{im\theta}$, so it is not the spin-j image of any rotation (our reading of [27]).

2.4 Two roles

Each application below uses the map in one of two ways.

Role I: coordinates on a space of states. The map relabels points of a space on which a group acts. Examples are Gaussian beams, squeezed states, loads on a transmission line and reflection coefficients of a thin film. The group ($\mathrm{SL}(2,\mathbb{R})$ in a "real" description, $\mathrm{SU}(1,1)$ in a "conjugate-pair" description) acts by Möbius maps in either picture.

Role II: on a generator or spectral variable. The map is applied to a quantity that lives on the real line when the system is lossless: a frequency, an energy, a reactance, a Hermitian operator, a reaction matrix. It produces something on the unit circle: a delay z, a one-step propagator, a reflection coefficient, an S-matrix.

The two roles are the same Möbius map. Role II is Role I applied to the spectrum. Functional calculus applies the scalar map (2) eigenvalue by eigenvalue, so a Hermitian operator, whose spectrum lies on the boundary $\mathbb{R}$, goes to a unitary one, whose spectrum lies on the circle. Some problems use both roles at once. Allen and Healy's treatment of broadband matching is one: the load is a point of the disk at each frequency (Role I), and its frequency dependence is an H^∞ function obtained by a Cayley map of the frequency half-plane (Role II) [28]. Wave digital filters are another (§5.1). Table 1 sorts the examples of this note.

2.5 Why the angle doubles

There are three equivalent ways to see it.

1. *Ratios of a conjugate pair.* By (7), a rotation by α multiplies the two members of the pair by $e^{i\alpha}$ and $e^{-i\alpha}$. Any ratio of the two therefore picks up $e^{2i\alpha}$. Each concrete double angle in this note is a ratio of this kind:

   - reflected over incident voltage wave on a line;
   - outgoing over incoming spherical wave in scattering;
   - E_-/E_+ in polarization;
   - a_+ over $a_-^\dagger$ in the two-photon formalism;
   - the ratio c of the $a^\dagger$ and a coefficients in the annihilator $a + c\,a^\dagger$ of a squeezed state (§3).

example	half-plane side (additive, "real")	disk side (multiplicative, "conjugate pair")
Role I: space of states		
Gaussian beams	complex beam parameter q, Im fixed sign	beam in Poincaré disk
squeezed vacua	width κ, $\operatorname{Re}\kappa > 0$	$c = e^{i\theta}\tanh r$, $\lvert c\rvert < 1$
quadrature directions	slope $u_1/u_2 \in \mathbb{R}$	$\chi(u)$, $\lvert\chi\rvert = 1$
polarization	Jones ratio E_y/E_x	circular ratio E_-/E_+
loads, thin films	impedance Z, $\operatorname{Re} Z > 0$	reflection Γ, $\lvert\Gamma\rvert < 1$
Role II: generator / spectral variable		
discretization	s (Laplace), A Hurwitz	z, A_d Schur stable
time stepping	$H\Delta t$ Hermitian	Crank–Nicolson U, unitary
scattering	reaction matrix K, Hermitian	S, unitary
self-adjoint extensions	symmetric operator	isometry; extensions $\leftrightarrow U(n)$
lossless cavity	detuning $\xi \in \mathbb{R}$	reflectivity, $\lvert r\rvert = 1$
random matrices	Cauchy (Lorentzian) ensemble	circular ensemble

Table 1: The two roles of the Cayley transform. In every row the boundary of the half-plane (a real quantity) corresponds to the unit circle (a pure phase), and the interior on one side corresponds to the interior on the other.

2. *Lines versus vectors.* A Möbius map acts on lines through the origin, not on vectors. The rotations by α and $\alpha + \pi$ act identically on lines, so the isometry group of the disk is $\mathrm{PSL}(2,\mathbb{R}) = \mathrm{SL}(2,\mathbb{R})/\{\pm 1\}$. A rotation through α in $\mathrm{SL}(2,\mathbb{R})$ rotates the disk by 2α [9]. The sign ambiguity of a homodyne direction ($\chi(-u) = \chi(u)$) is this quotient.

3. *Inscribed angles.* Figure 1c, or the half step forward and half step back of §2.3.

In the literature the half-angle appears separately in each field:

- Cayley's $\tan\frac{1}{2}\theta$ [14];
- Gibbs's semi-tangent [16];
- the spectral-theorem substitution $t = -\cot(s/2)$ in von Neumann's theory [29, §67];
- $K = \tan\delta$ with $S = e^{2i\delta}$ in scattering;
- the prewarping $\Omega = \frac{2}{T}\tan(\Omega_d T/2)$ in DSP [30];
- "360° corresponds to one-half wavelength" on the Smith chart [31];
- $\varphi = 2\arctan E$ in random-matrix theory [32].

We found no source that treats these as a single phenomenon.

3 Role I: Gaussian states, quadratures, and polarization

3.1 Gaussian states and Gaussian beams on one disk

Paraxial optics is the quantum mechanics of a free particle, with $\lambda/2\pi$ playing the role of $\hbar$ and the axial distance playing the role of time [2]. A Gaussian beam $\exp(ikx^2/2q)$ is the state

$\psi \propto e^{i\tau x^2/2}$ with $\tau = k/q$. In the convention of [2] the beam parameter has $\operatorname{Im} q < 0$, so τ lies in $\mathbb{H}$. A real ray matrix acts on τ by a Möbius map.

The disk picture comes from the ladder operators (5). Chi, Fraas and Tan define a squeezed state by [33]

$$(a + c\,a^\dagger)\,\psi_c = 0, \qquad \psi_c(x) \propto \exp\Big(-\frac{1+c}{1-c}\,\frac{x^2}{2}\Big), \tag{11}$$

which is normalizable exactly when $|c| < 1$. The width parameter $\kappa = (1+c)/(1-c)$ is the Cayley transform of the disk coordinate; equivalently $\tau = i\kappa$ gives $c = \mathcal{C}(\tau)$, with no extra rotation. For the standard squeezed vacuum $S(\xi)|0\rangle$, $\xi = re^{i\theta}$, one has $c = e^{i\theta}\tanh r$, the Perelomov SU(1, 1) coherent-state label [34]. (We checked this independently by building $S(\xi)|0\rangle$ in a truncated Fock space.)

- *Dynamics.* Quadratic Hamiltonians act on c by disk automorphisms. A phase-space rotation $e^{-i\phi a^\dagger a}$ sends $c \mapsto e^{-2i\phi}c$, the double angle again.
- *Classification.* The stable, free and unstable oscillators are the elliptic, parabolic and hyperbolic Möbius classes [33]. In optics these are the stable, marginal and unstable resonators [2, 35].

Distances are physical. The overlap of two pure Gaussian states depends only on their hyperbolic distance d [36]:

$$|\langle\psi_1|\psi_2\rangle|^2 = \operatorname{sech}\frac{d}{2} \quad \text{(one mode)}, \qquad 1 - \operatorname{sech}^2\frac{d}{2} = \left|\frac{\tau_1 - \tau_2}{\tau_1 - \bar\tau_2}\right|^2 = \left|\frac{c_1 - c_2}{1 - \bar c_1 c_2}\right|^2. \tag{12}$$

The right side is the squared pseudo-hyperbolic distance. For circular two-dimensional beams it is exactly the mode-mismatch formula of laser interferometry [37]. The disk form shows that it is a Möbius map that moves c_2 to the origin, evaluated at c_1.

3.2 The boundary: homodyne directions

What happens as $|c| \to 1$? Write $c = e^{2i\varphi}$. Then

$$a + e^{2i\varphi}a^\dagger = \sqrt{2}\,e^{i\varphi}\,(\cos\varphi\,\hat x + \sin\varphi\,\hat p). \tag{13}$$

The "state" annihilated by this operator is the eigenstate, with eigenvalue zero, of the quadrature $\hat X_\varphi = u \cdot \hat r$ with $u = (\cos\varphi, \sin\varphi)$. It is an infinitely squeezed state that a homodyne detector at angle φ would measure, and its boundary point is

$$c = e^{2i\varphi} = \chi(u). \tag{14}$$

Observation 1. The space of pure single-mode Gaussian states is the open disk $|c| < 1$. Its ideal boundary is the circle of homodyne directions, and the boundary point of the direction u is exactly the double-angle phasor $\chi(u)$ of (1). Anchors 1 and 2 are therefore the interior and the boundary of one disk. On the optical side, the boundary consists of real rays: point sources and plane waves [2]. In multilayer optics it consists of perfect mirrors [38].

We did not find Observation 1 stated in the squeezed-state literature we read [8, 33, 34], although each step is elementary.

3.3 Quadratures and sidebands

The two-photon formalism of Caves and Schumaker [3] builds quadratures from a pair of sidebands at $\omega_0 \pm \Omega$. In the notation of KLMTV [4, eq. 6],

$$a_1 = \frac{a_+ + a_-^\dagger}{\sqrt{2}}, \quad a_2 = \frac{a_+ - a_-^\dagger}{\sqrt{2}\,i} \qquad \Longleftrightarrow \qquad \begin{pmatrix} a_+ \\ a_-^\dagger \end{pmatrix} = \Omega \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}. \tag{15}$$

The conversion from quadratures to sidebands is the Cayley matrix (5). Kwee *et al.* write the transfer matrix of a filter cavity as this conversion applied to the diagonal matrix of sideband reflectivities [39, eqs. 11–12]:

$$T_{\mathrm{fc}} = \Omega^{-1}\,\mathrm{diag}(r(\Omega),\, \bar{r}(-\Omega))\,\Omega. \tag{16}$$

A lossless cavity applies phases $e^{i\alpha_\pm}$ to the two sidebands. So χ, the ratio of the a_+ and $a_-^\dagger$ components, picks up $e^{i(\alpha_+ + \alpha_-)}$. The quadratures are rotated by $\alpha_p = \frac{1}{2}(\alpha_+ + \alpha_-)$ and multiplied by a common phase $e^{i\alpha_m}$, $\alpha_m = \frac{1}{2}(\alpha_+ - \alpha_-)$ [4, eq. 79].

Each sideband phase is itself a Cayley transform. Kwee *et al.* write the reflectivity of a high-finesse cavity with fractional loss ϵ as $r = (\epsilon - 1 + i\xi)/(1 + i\xi)$, where ξ is the detuning normalized to the linewidth [39, eq. 7]. In the lossless case $\epsilon = 0$ this is $r = -(1 - i\xi)/(1 + i\xi)$: the real detuning axis maps onto the unit circle, with phase $\pi - 2\arctan\xi$. With loss, the image is a circle *inside* the disk. Purdue and Chen take the step that turns a design condition on angles into one on polynomials. They write $e^{2i\zeta} = (1 + i\tan\zeta)/(1 - i\tan\zeta)$ and expand [40, App. A]. A sum of filter-cavity angles becomes a product of first-order all-pass factors. The companion note [5] makes this a pole-placement theorem; we return to it in §5.3.

3.4 Polarization: the compact twin

The same construction on a field (E_x, E_y) is the change from the linear to the circular basis, $E_\pm = (E_x \mp iE_y)/\sqrt{2}$. Then

$$\frac{E_-}{E_+} = \frac{E_x + iE_y}{E_x - iE_y} = \chi(E), \qquad \text{and} \qquad \frac{E_-}{E_+} = \frac{1 + i\zeta}{1 - i\zeta} \quad \text{with} \quad \zeta = \frac{E_y}{E_x}. \tag{17}$$

So the change of polarization basis is a Cayley transform of the Jones ratio. Bliokh, Alonso and Dennis identify E_-/E_+ as the stereographic coordinate of the Poincaré sphere [41, Sec. IV B], whose azimuth is twice the orientation angle of the ellipse. For a linearly polarized field, (E_x, E_y) is real and $E_-/E_+ = e^{2i\psi}$ lies on the equator. KLMTV's χ of a quadrature direction is precisely the circular-basis ratio of a linear polarization. In 1972 Azzam and Bashara showed that the Jones ratio obeys a Riccati equation (the infinitesimal form of a Möbius map) in anisotropic media and is carried to the Poincaré sphere by stereographic projection [42]. Han, Kim and Noz describe Jones matrices acting on it as a bilinear representation of the Lorentz group [43].

Polarization is the *compact* case. Lossless polarization optics preserves the intensity $|E_+|^2 + |E_-|^2$, so the group is SU(2) acting on a sphere, and the Cayley map is a rotation of that sphere. Gaussian states and lossless scattering preserve a *difference*, $|a|^2 - |b|^2$, so the group is SU(1, 1) acting on a disk. Yurke, McCall and Klauder classify interferometers by exactly this distinction: passive beam-splitter interferometers are SU(2) and parametric-amplifier interferometers are SU(1, 1) [44]. Deschamps, from the 1950s on, described transmission lines, waveguide junctions and polarization by one geometrical model [45, 46].

4 Role II: from continuous to discrete time

4.1 Tustin's map

Replacing the integrator $1/s$ by the trapezoidal rule $y_n - y_{n-1} = \frac{T}{2}(u_n + u_{n-1})$ gives Tustin's substitution [6, 47]:

$$z = \frac{1 + sT/2}{1 - sT/2} = \operatorname{cay}\Big(\frac{sT}{2}\Big), \qquad z\big|_{s=j\Omega} = e^{\,2j\arctan(\Omega T/2)}. \tag{18}$$

It is the $(1,1)$ Padé approximant of e^{sT}. It maps the stable left half-plane onto the unit disk and the frequency axis onto the unit circle, wrapping it exactly once. The frequency warping $\Omega_d T = 2\arctan(\Omega T/2)$, which designers undo by "prewarping" a critical frequency [30, 48], is the half-angle (4) (Fig. 1d). The textbooks we checked present the warping as a design nuisance and do not connect it with the hyperbolic geometry of §3.

4.2 The state-space map is exact

The bilinear map changes more than transfer functions. With $\alpha = 2/T$, define

$$\begin{aligned} A_d &= (\alpha + A)(\alpha - A)^{-1}, \quad & B_d &= \sqrt{2\alpha}\,(\alpha - A)^{-1}B, \\ C_d &= \sqrt{2\alpha}\,C(\alpha - A)^{-1}, \quad & D_d &= D + C(\alpha - A)^{-1}B. \end{aligned} \tag{19}$$

Then $G_d(z) = G(s)$ exactly, and the controllability Gramians coincide:

$$AP + PA^T + BB^T = 0 \quad \Longleftrightarrow \quad A_d P A_d^T - P + B_d B_d^T = 0. \tag{20}$$

Proof. Let $F = (\alpha - A)^{-1}$, which commutes with A. Then $A_d P A_d^T - P = F\big[(\alpha + A)P(\alpha + A)^T - (\alpha - A)P(\alpha - A)^T\big]F^T = 2\alpha F(AP + PA^T)F^T = -B_d B_d^T$. □

The same holds for the observability Gramian. As a result:

- Hankel singular values and balanced realizations carry over unchanged.
- The H_∞ norm is preserved, because the frequency axis maps bijectively onto the circle.
- The McMillan degree is preserved.
- By Proposition 1, the Hamiltonian matrix of a continuous-time Riccati equation maps to a symplectic matrix with the same invariant subspaces. The stabilizing solution of the continuous-time Riccati equation therefore equals that of the corresponding discrete-time one.

Laub used this to reuse modulus-ordered Schur solvers [49]; see also [50]. We verified all of these numerically, but did not find them collected as one theorem in the sources we could read.

4.3 Crank–Nicolson is Tustin for the Schrödinger equation

Apply (18) to $i\hbar\dot{\psi} = H\psi$. The result is the Crank–Nicolson step,

$$U_{\mathrm{CN}} = \Big(1 + \frac{iH\Delta t}{2\hbar}\Big)^{-1}\Big(1 - \frac{iH\Delta t}{2\hbar}\Big), \tag{21}$$

which *Numerical Recipes* calls "Cayley's form" [51]. It was famously used for the wave-packet movies of Goldberg, Schey and Schwartz [52]. By Proposition 1 it is exactly unitary. It commutes with H, so energy is conserved exactly [53]. Its only error is in the phase: an eigenstate of energy E advances by $2\arctan(E\Delta t/2\hbar)$ per step instead of $E\Delta t/\hbar$. The phase error per step is $2\arctan(x/2) - x = -x^3/12 + O(x^5)$ with $x = E\Delta t/\hbar$. This is the same warping as Tustin's, now in energy rather than frequency. Higher-order unitary steppers built from the diagonal $[M/M]$ Padé approximant of e^z factor into M Cayley factors [19, 20]. So they are finite *Blaschke products*: rational functions that are unimodular on the boundary. We meet Blaschke products again next.

5 Both roles at once: impedance and scattering

5.1 The Smith chart and passive networks

For a load of normalized impedance $\zeta = Z/Z_0$ on a line, the reflection coefficient is

$$\Gamma = \frac{\zeta - 1}{\zeta + 1} = \mathcal{C}(i\zeta). \tag{22}$$

This maps passive loads, $\operatorname{Re}\zeta > 0$, onto the disk. The Smith chart [54] is this disk with the impedance grid drawn on it. Smith later recalled that the decisive hint came from noticing that the circles of the rectangular impedance chart "correspond exactly to the lines of force and the equipotentials" of two line charges. He also recalled that a "bilinear conformal transformation" then gave the chart [31]. A length l of lossless line rotates Γ by $e^{-2j\beta l}$. The forward and backward waves pick up opposite phases, so their ratio doubles the angle: "360° corresponds to one-half wavelength" [31].

The chart is the Poincaré disk, and this was understood early. Deschamps (1953–54) and Weissfloch (1954) treated lossless two-ports as Möbius maps of the reflection disk [45, 55]. By 1970 the engineering handbook stated that a lossless junction "leaves invariant the hyperbolic distance" and that "the hyperbolic distance to the origin of the chart is the mismatch" (quoted in [28]). The dictionary, made precise by Helton's "non-Euclidean" program [56] and by Allen and Healy [28], is:

- $|\Gamma| = \tanh(d/2)$ and $\mathrm{VSWR} = e^d$, where d is the hyperbolic distance from the matched load;
- the power delivered from a source Z_S to a load Z_L is

$$\frac{P_L}{P_{\text{avail}}} = \frac{4R_S R_L}{|Z_S + Z_L|^2} = 1 - \left|\frac{Z_L - \bar{Z}_S}{Z_L + Z_S}\right|^2 = \operatorname{sech}^2 \frac{d(Z_L, \bar{Z}_S)}{2}. \tag{23}$$

Conjugate matching means zero distance.

Compare (23) with (12). The power-transfer theorem of circuit theory and the mode-matching efficiency of laser optics are *the same formula*: one minus a squared pseudo-hyperbolic distance. The reason is the same in both: a unitary overlap is invariant under the group, and the only invariant of a pair of points is their distance. (We did not find the two formulas compared in the literature.)

The scattering matrix of an n-port is the matrix Cayley transform of its normalized impedance matrix, $S = (Z - I)(Z + I)^{-1}$ [57, 58]. Passivity becomes bounded-realness through the matrix version of (3),

$$I - S^\dagger S = 2(Z + I)^{-\dagger}(Z + Z^\dagger)(Z + I)^{-1}, \tag{24}$$

which links the positive-real and bounded-real lemmas [59]. Losslessness becomes unitarity on the frequency axis, and lossless lumped circuits are *Blaschke products* [28]. In discrete time, all-pass filters are finite Blaschke products whose phase falls by $M\pi$ over half the circle, and they are realized as cascades of first-order sections [60]. Peeling off one first-order Cayley factor at a time is the Schur–Cohn stability test in discrete time and Routh's test in continuous time [61].

Both maps at once: wave digital filters. Fettweis's wave digital filters [62] combine both roles. Rewrite a lossless LC network in wave variables: this is Role I, with reflectance $(Z - R)/(Z + R)$. Then apply the bilinear transform: this is Role II. With port resistance $R = T/2C$ for a capacitor and $R = 2L/T$ for an inductor, the two Möbius maps are inverse to each other and cancel. A capacitor becomes the pure delay z^{-1} and an inductor becomes $-z^{-1}$ [48]. Passivity of the analog circuit becomes structural passivity of the digital filter.

5.2 The K-matrix and the S-matrix

In single-channel scattering, unitarity forces $S = e^{2i\delta}$, and the reaction matrix is $K = \tan\delta$:

$$S = \frac{1 + iK}{1 - iK} = \frac{1 + i\tan\delta}{1 - i\tan\delta} = e^{2i\delta}. \tag{25}$$

The phase shift δ is literally the half-angle of (4). Heitler's reaction matrix [63] works with a Hermitian K, which only needs principal-value propagators, and restores unitarity exactly through this Cayley form. Landshoff's 1968 paper is titled "Cayley transform of the S matrix" [64]. For resonance scattering, an internal Hamiltonian H ($N \times N$) coupled to M channels by W gives the Mahaux–Weidenmüller form [65–67]

$$S(E) = \frac{1 - iK}{1 + iK}, \qquad K(E) = \pi W^\dagger (E - H)^{-1} W. \tag{26}$$

(Some authors write $(1 + iK)/(1 - iK)$ with $K \to -K$; Appendix A.) Here K is a matrix Pick (Herglotz) function of E, up to sign. Wigner proved in 1951 that collision-theory response functions have exactly these analytic properties [68]. By the determinant lemma $\det(1 + iK) = \det(E - H + i\pi WW^\dagger)/\det(E - H)$, so

$$\det S(E) = \prod_{n=1}^{N} \frac{E - \bar{z}_n}{E - z_n}, \qquad z_n \in \operatorname{spec}\left(H - i\pi WW^\dagger\right),\ \operatorname{Im} z_n < 0. \tag{27}$$

The S-matrix is a finite Blaschke product whose poles are the resonances. Its phase winds once around the circle for each resonance, since δ advances by π. Fyodorov and Sommers use this as an exact level-counting identity [66, eq. 42]. It is a finite-dimensional cousin of Levinson's theorem [69], and the scattering analogue of the all-pass phase winding in [60].

Three bridges to the rest of this note are explicit in the literature.

1. Fyodorov and Savin point out that $Z = iK$ "has the meaning of the normalized cavity impedance." So (26) is the microwave reflection formula, and they place it in the input–output theory of open systems going back to Livšic [70].

2. In the SLH description of passive quantum networks, the transfer function $\Xi(s) = D - C(sI - A)^{-1}C^\dagger D$ with $A = -iH_{\text{eff}}$ has the same structure [71]. A single cavity mode gives $\Xi(s) = (s - \kappa/2)/(s + \kappa/2)$, one Cayley factor. Cascading systems multiplies transfer functions [72].

3. Gough proves that a quantum transmission line's $S = (I - \Sigma)/(I + \Sigma)$ is lossless bounded real exactly when Σ is lossless positive real [73]. That is the circuit theorem (24), stated for quantum systems.

5.3 The KLMTV filter is a K-matrix problem

We can now return to anchor 2. In KLMTV's input–output relation the back-action enters through the lower-triangular matrix $\left(\begin{smallmatrix} 1 & 0 \\ -\mathcal{K} & 1 \end{smallmatrix}\right)$. In ray optics this is a thin lens of power $\mathcal{K}$. The back-action-free (variational) readout measures the direction $u \propto (\mathcal{K}, 1)$ [4, 5]. In lens language, u reads out position one focal length downstream, where position depends only on the input angle. Requiring a chain of lossless filter cavities to rotate a fixed homodyne angle onto this direction at every sideband frequency gives, in the conventions of [5], the target all-pass

$$\mathfrak{A}(s) = \frac{1 + i\mathcal{K}}{1 - i\mathcal{K}} = \frac{D + iN}{D - iN}, \qquad \mathcal{K} = \frac{N(s)}{D(s)}, \quad s = \Omega^2. \tag{28}$$

One lossless detuned cavity contributes one Blaschke factor $(s - \bar{p})/(s - p)$ in $s = \Omega^2$, so the number of cavities equals the degree of $\mathcal{K}$ [5].

Observation 2. Equation (28) is Heitler's map $S = (1 + iK)/(1 - iK)$, with the optomechanical coupling $\mathcal{K}$ in the role of the reaction matrix. The dictionary with §5.2 is:

- the filter cavities are the resonances, and #cavities = $\deg \mathcal{K}$ is #resonances = $\dim H$;
- the zeros of D (here the arm-cavity poles $s(s + \gamma^2)$) are the *bare* levels where K has real poles;
- coupling to the continuum pushes these bare levels off the real axis into the roots of $D - iN$. This is why the arm-cavity poles drop out of $\mathfrak{A}$.

Scattering theory computes S from given resonances. Filter synthesis is the inverse problem: find the resonances (cavities) that realize a given K. The algebra is identical.

We did not find this connection in the gravitational-wave literature we read. The closest is Purdue and Chen's use of $(1 + i\tan)/(1 - i\tan)$ to reduce the design to polynomial root matching [40].

5.4 Random matrices: Lorentzian is the shadow of uniform

If a phase φ is uniform on the circle, then $E = \tan(\varphi/2)$ has the Cauchy (Lorentzian) density $1/\pi(1 + E^2)$. The matrix version is due to Brouwer [32]. If H is drawn from the "Lorentzian ensemble," then $S = (1 + iH)/(1 - iH)$ is exactly distributed according to Dyson's circular ensemble, with eigenphases $\varphi_j = 2\arctan E_j$. Applied to the K-matrix, the same map carries Cauchy-distributed reaction matrices to the Poisson kernel $P(S) \propto |\det(1 - \bar{S}^\dagger S)|^{-(\beta N + 2 - \beta)}$, the maximum-entropy distribution of Mello, Pereyra and Seligman [67, 74]. It is the matrix version of harmonic measure on the disk, and its name and form go back to Hua [13]. Witte and Forrester use the same projection, written $\lambda = \cot(\theta/2)$, to move results between the Cauchy and circular Jacobi ensembles [75].

5.5 Boundary conditions: von Neumann's Cayley transform

Von Neumann built the theory of unbounded self-adjoint operators on the operator Cayley transform $V = (A - i)(A + i)^{-1}$ [29, 76]. The main facts are:

- A symmetric A gives an isometry V, and A is self-adjoint exactly when V is unitary.
- The self-adjoint extensions of A correspond to unitary maps between its deficiency subspaces.
- The spectral theorem passes from the circle to the line by $t = -\cot(s/2)$.

For physicists the content is that *boundary conditions form a unitary group*, and that the physically natural real parameter is the Cayley coordinate on it. Bonneau, Faraut and Valent work this out for teaching [10]. For the free particle on a half-line, the self-adjoint boundary conditions are

$$\varphi'(0) - i\varphi(0) = e^{i\alpha}(\varphi'(0) + i\varphi(0)) \quad \Longleftrightarrow \quad \varphi(0) = \lambda\varphi'(0), \quad \lambda = -\tan(\alpha/2), \tag{29}$$

a Robin condition whose parameter is a half-angle tangent [10, eq. 20]. A wave $Ae^{-ikx} + Be^{ikx}$ is then reflected with $B/A = -(1 + i\lambda k)/(1 - i\lambda k)$, a Cayley factor in λk. Dirichlet ($\lambda = 0$) and Neumann ($\lambda = \infty$) are antipodal on the circle of boundary conditions. Changing the boundary condition acts on the Weyl m-function, a Herglotz function, by a rotation in SO(2) written as a Möbius map, with the boundary condition labelled by $e^{2i\alpha}$ [77, eq. 7.59]. The general correspondence between Herglotz (or Carathéodory) and Schur functions, $F = (1 + f)/(1 - f)$, is classical; see Simon's survey [78].

6 Shorter appearances

Fresnel coefficients and multilayers. At an interface, the reflection coefficient $\rho = (\eta_2 - \eta_1)/(\eta_2 + \eta_1)$ is the Cayley transform of an impedance ratio. Through a stack of lossless layers, Γ obeys a disk-automorphism recursion while the wave impedance obeys a real Möbius recursion on the half-plane [79]. Vigoureux noticed that reflection coefficients compose by a complex version of Einstein's velocity addition [80]. Monzón and Sánchez-Soto made the identification of lossless stacks with $\mathrm{SU}(1,1) \simeq \mathrm{SO}(2,1)$ precise [81]. Yonte *et al.* use a single unitary that both conjugates the SU(1, 1) transfer matrix into an $\mathrm{SL}(2,\mathbb{R})$ "ABCD" matrix and maps the reflection disk onto the half-plane [82]. Orfanidis's equal-thickness recursion uses $s = (1 - z^{-1})/(1 + z^{-1})$, which equals $j \tan\delta$ on the circle: Tustin's map again, now in optics [79].

Velocities and rapidities. In one dimension, Einstein's addition $(v_1 + v_2)/(1 + v_1 v_2)$ is a disk automorphism fixing the real diameter. The real Cayley map $v \mapsto (1 + v)/(1 - v)$ turns it into multiplication of squared Doppler factors $k^2 = e^{2\xi}$. The "relativistic half velocity" $w = \tanh(\xi/2)$ satisfies $v = 2w/(1 + w^2)$ [83], the hyperbolic twin of $\tan\theta = 2t/(1 - t^2)$.

Spacecraft attitude. The Gibbs vector $\hat{n}\tan(\theta/2)$ and the "modified Rodrigues parameters" $\hat{n}\tan(\theta/4)$ are first- and second-order Cayley transforms. Each step halves the angle again and pushes the singularity further away [84].

7 Why the Cayley transform keeps appearing

With the examples in hand, we can say what they share. The map appears when three ingredients are present.

1. **A conserved quadratic quantity of signature** $(1,1)$**.** Examples are the symplectic form of ray optics and of the quadratures, energy flux on a line ($|V^+|^2 - |V^-|^2$), probability current (in minus out), the Wronskian of a boundary value problem, and the Bogoliubov constraint $|\alpha|^2 - |\beta|^2 = 1$. In a *real* basis (position and angle, x and p, V and I, standing waves) the quantity is a skew form. In a *conjugate-pair* basis (a and $a^\dagger$, forward and backward waves, outgoing and incoming waves, sidebands, circular components) it is a difference of norms. The identity (9) is the change of basis, and Ω is its matrix. "Real" descriptions are additive and natural for writing down physics: Hamiltonians, impedances in series, reactances, K-matrices. "Conjugate-pair" descriptions are multiplicative and natural for composing and measuring: cascades multiply, and reflection and transmission are ratios.

2. **Observables are ratios.** Wavefront curvature x/θ, impedance V/I, reflection V^-/V^+, a homodyne direction, a polarization state and a phase shift are all defined only up to an overall factor. Linear maps act on such ratios by Möbius maps. Since a ratio of two members of a conjugate pair picks up the *difference* of their phases, the angles double (§2.5).

3. **Losslessness lives on the boundary.** In each case the energy-conserving objects are the real ones on the half-plane side (real rays, reactances, Hermitian K, real frequency) and unimodular ones on the disk side (pure phases, unitary S, all-pass filters). Dissipation, normalizability and passivity become "inside the disk," by (3) and (24). In the frequency domain, causality and losslessness force inner functions. Rational inner functions are finite Blaschke products, and their degree counts resonances, cavities or windings.

In this language Tustin's map and Crank–Nicolson come from ingredient 3, applied to the generator of time translation. "Half a step forward equals half a step back" is Cayley's 1858 midpoint [17] and Gibbs's $\mathbf{c} \mp \mathbf{g} \times \mathbf{c}$, and it produces the same half-angle.

When not to expect it.

- The groups must be quadratic. The Cayley map has no special status for groups that do not preserve a bilinear form.
- It is not compatible with tensor products or higher representations (§2.3).
- It cannot reach eigenvalue -1, for example the half-wave plate or the quarter-period of an oscillator, without a change of base point.
- Lossy elements give contractions: the Gaussian aperture, the lossy cavity of Kwee *et al.*, the absorbing cavity with S sub-unitary. These form a semigroup that maps the disk into itself, not a group of isometries [39, 70].

8 Summary

appearance	what the Cayley transform does
Gaussian beam / squeezed state	half-plane width $\tau \to$ disk point $c = \mathcal{C}(\tau)$
homodyne direction	boundary point $\chi(u)$ of the same disk
quadratures $\to$ sidebands, $(x, p) \to (a, a^\dagger)$	the matrix Ω; $\mathrm{SL}(2,\mathbb{R}) \to \mathrm{SU}(1,1)$
linear $\to$ circular polarization	$\zeta \mapsto (1 + i\zeta)/(1 - i\zeta)$ on the Poincaré sphere
Smith chart	impedance $\to$ reflection; VSWR $= e^d$
power transfer, mode matching	$\operatorname{sech}^2(d/2)$ for both
Tustin / Crank–Nicolson	generator $\to$ one-step map; warping $2\arctan(x/2)$
Riccati, Lyapunov	Hamiltonian $\to$ symplectic; Gramians unchanged
K-matrix $\to$ S-matrix	$e^{2i\delta} = (1 + i\tan\delta)/(1 - i\tan\delta)$; Blaschke product
KLMTV filter cavities	inverse K-matrix problem; #cavities = #resonances
random matrices	Cauchy ensemble $\to$ circular ensemble
self-adjoint extensions	symmetric $\to$ isometric; Robin $\lambda = -\tan(\alpha/2)$
rotations (Cayley 1846)	skew $\to$ orthogonal; Gibbs vector $\hat{n}\tan(\theta/2)$

A reading path

- *The map and its geometry:* Needham [85] or Anderson [86] for pictures; Casselman's notes [9] for exact formulas.
- *One survey across physics:* Sánchez-Soto *et al.* [7], on transfer matrices in quantum mechanics, optics and mechanics (Role I only).
- *Engineering:* Allen and Healy [28], §4, free, on the Smith chart as hyperbolic geometry; Regalia, Mitra and Vaidyanathan [60] on all-pass filters.
- *Squeezed states on the disk:* Chi, Fraas and Tan [33]; Arvind *et al.* [8], §4.
- *Scattering:* Fyodorov and Savin [70], §34.1; Brouwer [32], §III.
- *Boundary conditions:* Bonneau, Faraut and Valent [10].
- *History:* Cayley 1846 and 1858 [14, 17], both short; Gibbs–Wilson Art. 127 [16].

A Conventions across sources

Every form of "the" Cayley transform below is (2) composed with a rotation of the disk or of the half-plane.

source	formula	relation to $\mathcal{C}(z) = \frac{z-i}{z+i}$
Casselman; Hardy *et al.*; von Neumann	$(z-i)/(z+i)$	identical
half-angle form; Tustin; Brouwer	$(1+ix)/(1-ix)$	$-\mathcal{C}(x)$
Smith chart; Allen–Healy; Carlin	$(Z-1)/(Z+1)$	$\mathcal{C}(iZ)$ (right half-plane)
Arvind; KLMTV; Kwee (projectively)	$(t+i)/(t-i)$	$1/\mathcal{C}(t)$
Yonte *et al.*; Sánchez-Soto	$w = (z+i)/(1+iz)$, disk $\to \mathbb{H}$	disk coordinate $z = i\,\mathcal{C}(w)$
Kwee (lossless cavity)	$-(1-i\xi)/(1+i\xi)$	$1/\mathcal{C}(\xi)$
Fyodorov–Sommers; Mahaux–Weidenmüller	$(1-iK)/(1+iK)$	$-\overline{\mathcal{C}(K)}$ for real K; Brouwer's form with $K \to -K$
Chi–Fraas–Tan	$\kappa = (1+c)/(1-c)$	$c = \mathcal{C}(i\kappa)$
Akhiezer–Glazman; Witte–Forrester	$t = -\cot(s/2)$; $\lambda = \cot(\theta/2)$	base point moved by π

Time conventions also move poles between half-planes. With physics $e^{-i\omega t}$, resonance poles have $\operatorname{Im}\omega < 0$. With engineering $e^{j\omega t}$ and Laplace s, they have $\operatorname{Re} s < 0$. Kwee *et al.* place filter-cavity poles in the opposite half-plane from KLMTV and Purdue–Chen, and Purdue–Chen's eqs. (A12) and (A13) disagree in sign as printed. KLMTV's eq. (88) lacks a factor of 2, as Purdue and Chen point out [40].

B Numerical checks

All checks are in `scripts/verify_cayley.py`; the output is `scripts/verify_cayley.out`. All 30 pass, with errors at round-off level except where noted.

- *The map.* (4); (3) and its matrix form (24); $\Omega\,\mathrm{SL}(2,\mathbb{R})\,\Omega^{-1} = \mathrm{SU}(1,1)$ with rotations diagonal; the action $\chi(R(\alpha)u) = e^{2i\alpha}\chi(u)$.
- *Gaussian states.* The squeezed vacuum $S(\xi)|0\rangle$, built in a 160-level Fock space, has Gaussian exponent $\kappa = (1+c)/(1-c)$, $c = e^{i\theta}\tanh r$, to 10^{-15}. The rotation $e^{-i\phi N}$ multiplies c by $e^{-2i\phi}$.
- *Discretization.* (18) and (19): transfer functions, Gramians, Hankel singular values, and the stabilizing Riccati solution computed from the Hamiltonian and from its Cayley transform agree to 10^{-14}. Hamiltonian maps to symplectic.
- *Crank–Nicolson.* It is unitary, its eigenphases are $-2\arctan(E\Delta t/2)$, and the series $-x^3/12$ holds.
- *Scattering.* (26) equals the Blaschke product (27) to 10^{-13}. The phase winds once per resonance (6 resonances, winding 6.00).
- *Random matrices.* For a CUE matrix U, $i(1-U)(1+U)^{-1}$ is Hermitian, and its eigenvalue distribution matches the Cauchy law (Kolmogorov–Smirnov distance 3×10^{-3} over 32 000 eigenvalues).
- *Smith chart.* $|\Gamma| = \tanh(d/2)$, $\mathrm{VSWR} = e^d$, (23), and the rotation $e^{-2j\beta l}$.
- *Wave digital filters.* A capacitor gives z^{-1} and an inductor gives $-z^{-1}$.

- *Other.* The polarization ratio and the Stokes vector by inverse stereographic projection; the Gibbs-vector rotation; the Robin-wall reflection; the SLH single-cavity transfer function; the KLMTV target (28) as two Blaschke factors; the rapidity formulas.

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